Quantum computation in standard Qwen layers

Circuit instructions enter as tokens; model-generated digits carry the result.

a Task and request

Start in |000000⟩

Apply H3 → T3 → H3

Find P(q3 = 1)

Qwen

Circuit + indexed event

Host parses the request and loads instruction tokens.

b Shared model forward path

4 fixed-weight QPU layers

Gated DeltaNet

Amplitude digits

Program + counter

Fixed SwiGLU weights

H / X / T / CNOT

Born probabilities

Numerical token feedback → state writeback

Shared arithmetic at addressed amplitudes

32 Qwen language layers + model output head

c Numerical answer

Model output digits

↓ decode

P(q3 = 1)

0.1464464664

Independent reference

Qiskit: 0.1464466094

Absolute error: 1.43×10^{-7}

Six-qubit H–T–H calculation · GDN memory and gate arithmetic in the added layers